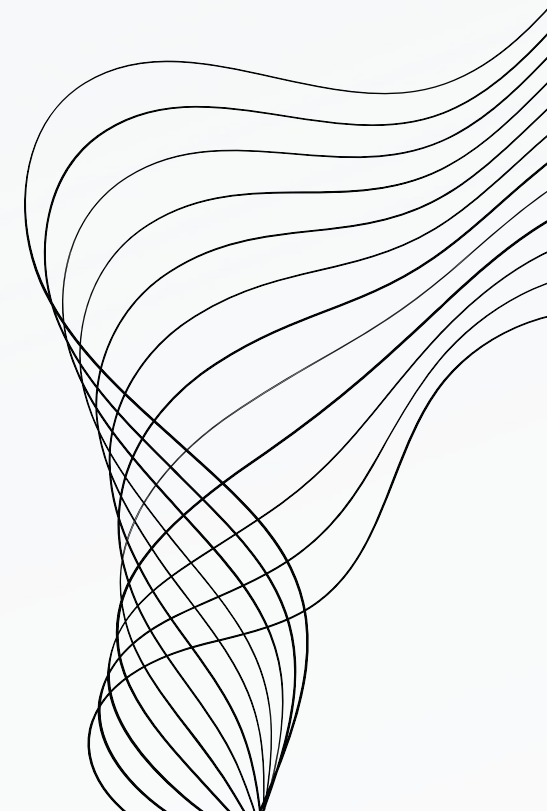
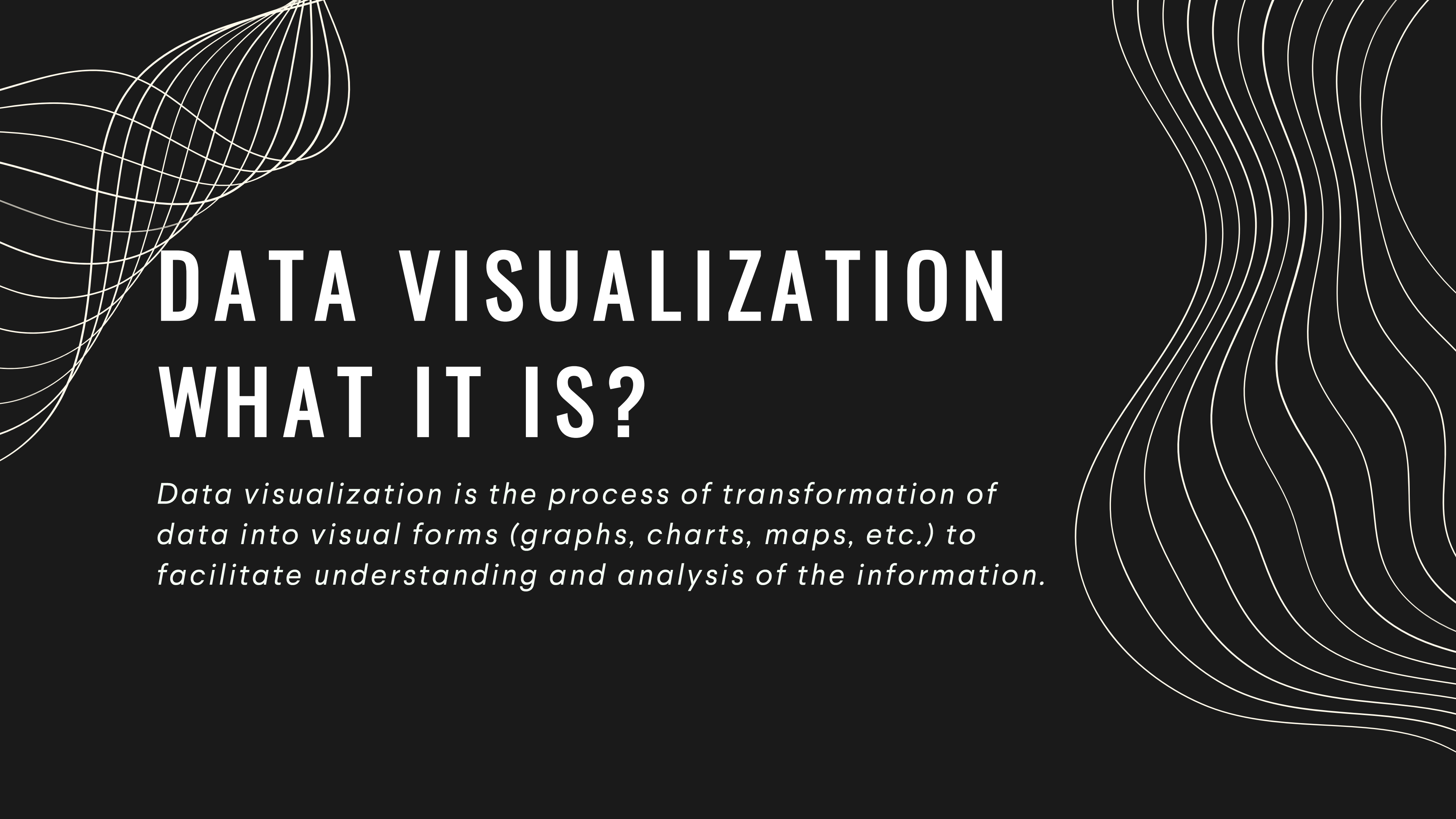


DATAVIS

LECTURE №1. INTRODUCTION TO VISUALIZATION





DATA VISUALIZATION

WHAT IT IS?

Data visualization is the process of transformation of data into visual forms (graphs, charts, maps, etc.) to facilitate understanding and analysis of the information.



SIGNS OF GOOD VISUALIZATION



01

MAKES DATA ACCESSIBLE

02

*COMBINES THE STRENGTHS OF HUMANS AND
COMPUTERS*

03

PROVIDES UNDERSTANDING

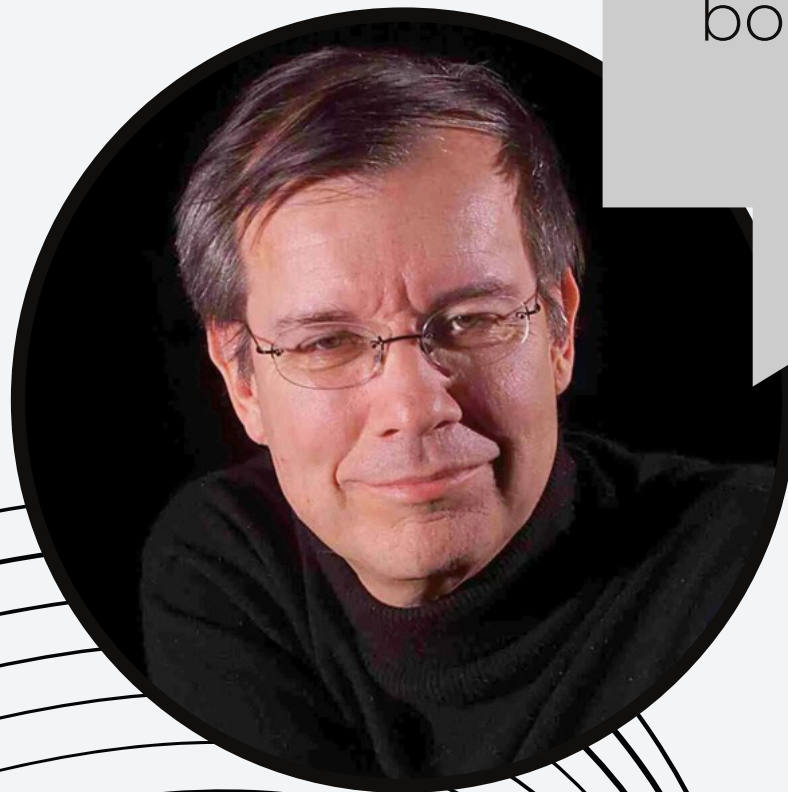
04

INTERACTS WITH THE VIEWER

VISUALIZATION

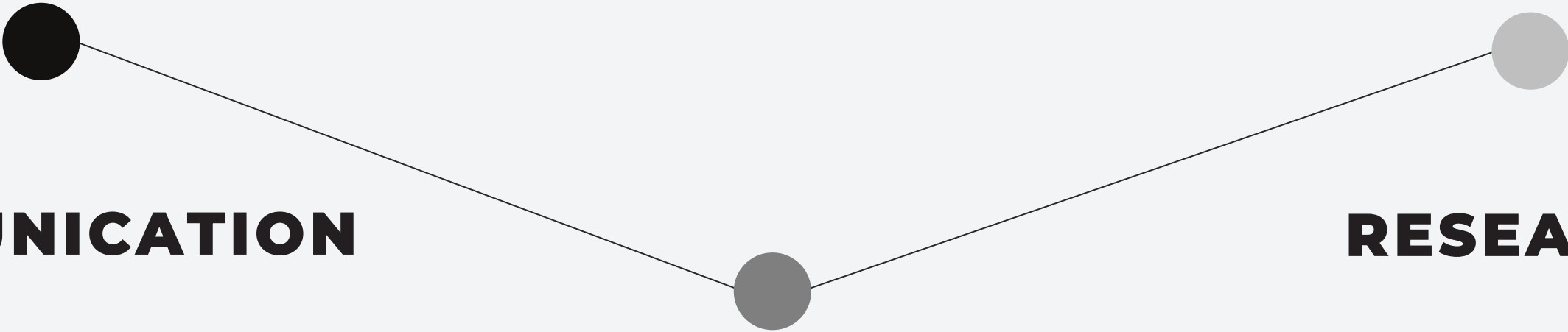
Stuart Card

«Visualization is really about external cognition, that is, how resources outside the mind can be used to boost the cognitive capabilities of the mind».



Author of the scientific bestseller “Readings in Information Visualization: Using Vision to Think”

USE CASES



COMMUNICATION

*Visualization can be used to make
information easier to understand*

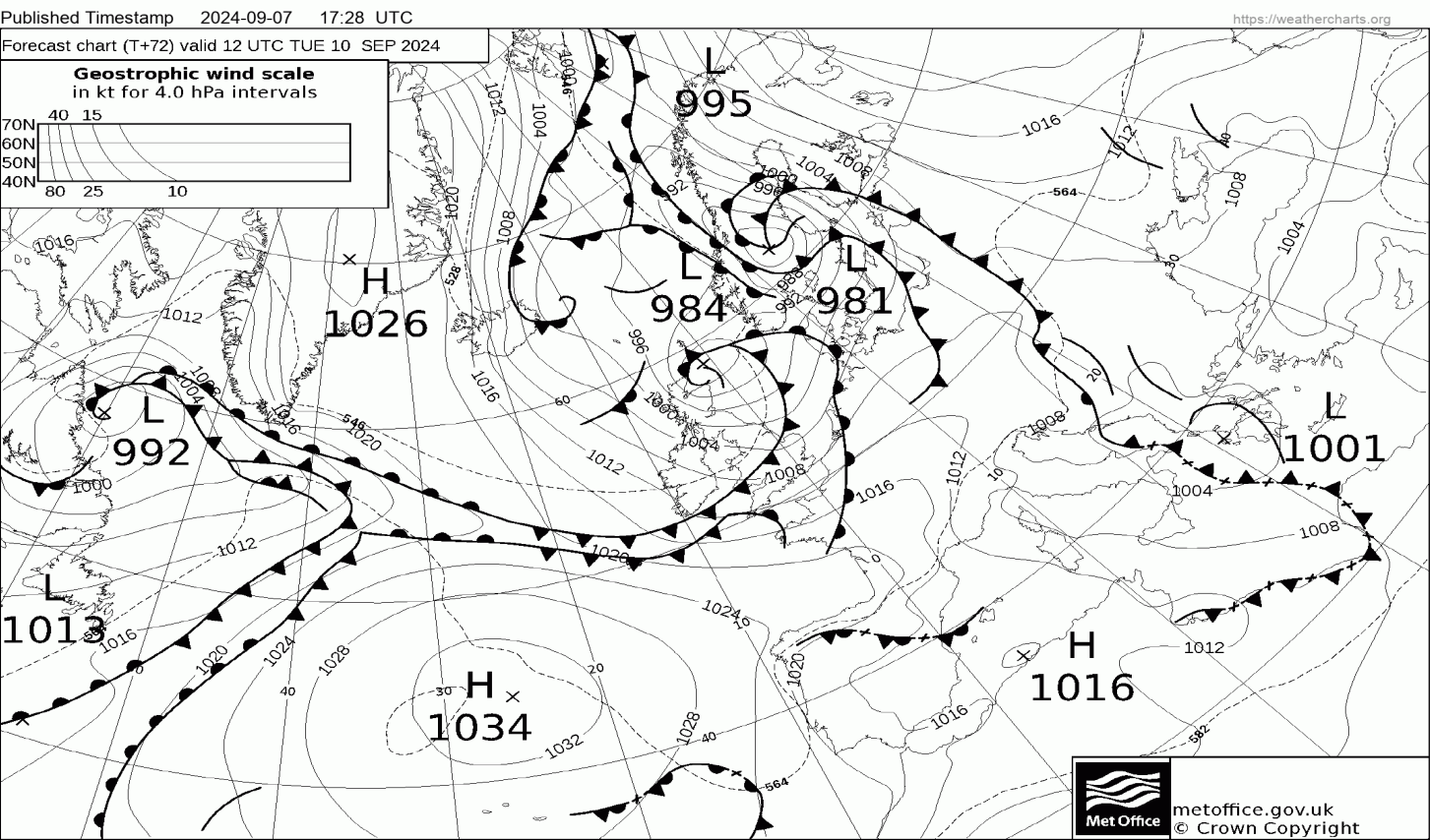
REINFORCEMENT

*Visualization can be used as
alternative sources to support
conclusions*

RESEARCH

*Visualization serves as an
exploratory tool, revealing hidden
connections*

RESEARCH: WEATHER DATA



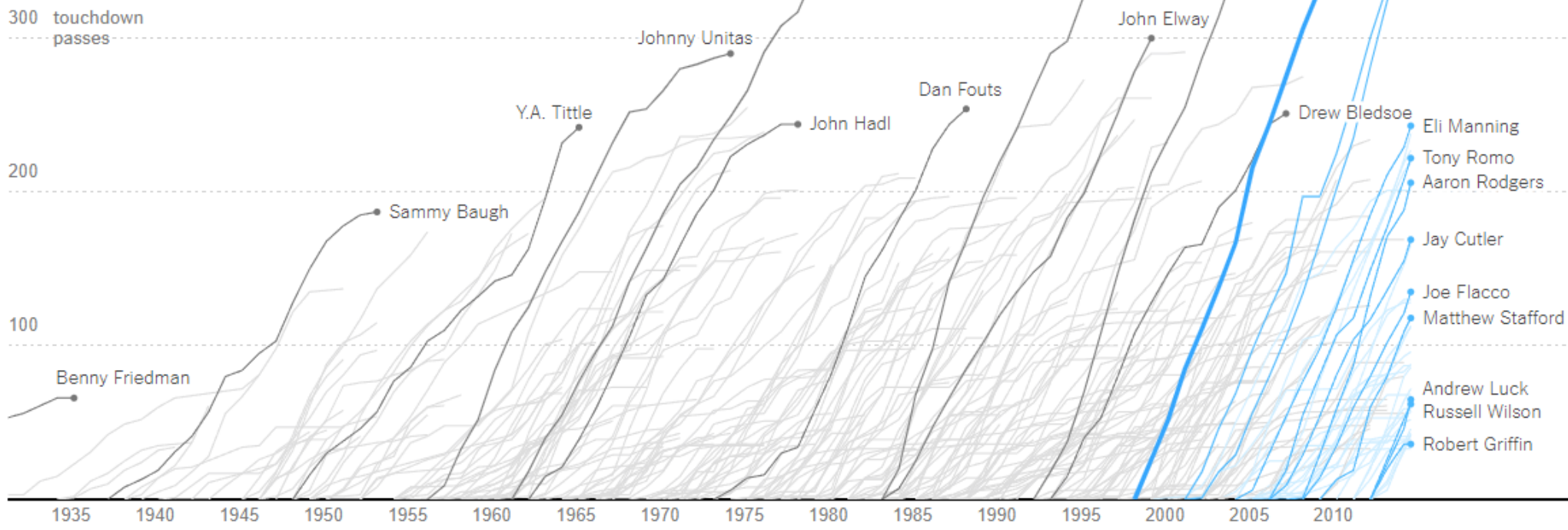
COMMUNICATION: DEMONSTRATION

Why Peyton Manning's Record Will Be Hard to Beat

By GREGOR AISCH and KEVIN QUEALY OCT. 19, 2014

The Broncos quarterback set the all-time N.F.L. touchdown passing record — and is still going strong.

RELATED ARTICLE





WHY USE GRAPHICS?

1. *Pictures are more **informative**. They contain more information in a smaller space;*
2. *Pictures create a “**gestalt**” effect: they give a general idea, making the structure more visual;*
3. *Pictures are more **accessible**. They are easier to understand, faster to learn, and they are better remembered.*



WHY USE COMPUTERS?

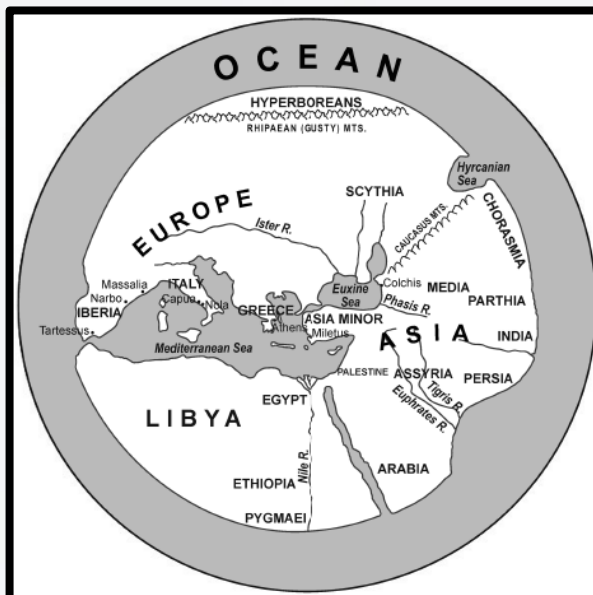
1. *Computers are more **efficient**;*
2. *They can draw graphics much **faster** than a person;*
3. *Computers can create **interactive** visualizations;*
4. *Data on computers can **be combined from different sources**;*
5. *Visualizations made on a computer can be **scaled** (expanded).*

Visualization in real life

<https://youtu.be/bIAF7kBbGKk>



FIRST VISUALIZATIONS



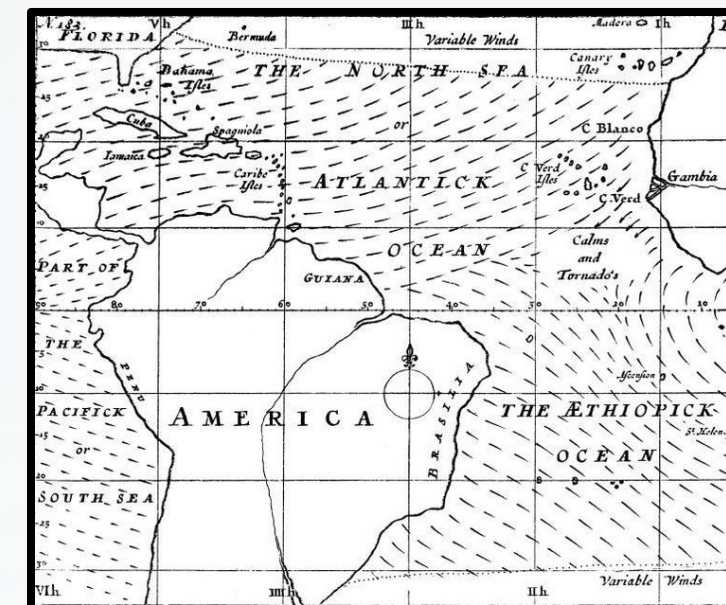
Data visualization is not a new phenomenon. People have used visual images to represent information since ancient times.

The purpose of the first visualizations was mainly limited to orientation (geography) and the transfer of knowledge between generations.

Considered the first map of the World (Earth) Anaximands of Miletus, ~550 BC

Over time, more advanced forms of visual images emerged and took root in people's minds, and today, just by looking at the visualization, we often already understand its meaning.

First weather map
Edmond Halley, 1686.

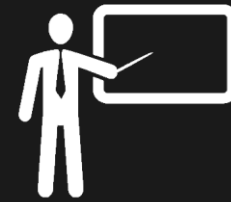


TYPES OF VISUALIZATIONS



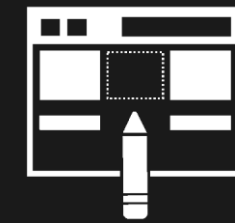
Schemes include graphs, diagrams, histograms, etc. They serve to display deterministic data in a convenient form.

SCHEMAS



Presentations, on the other hand, are used to structure and display non-deterministic information.

PRESENTATIONS



Dashboards are information panels for visualizing dynamically changing data and grouping it from different sources.

DASHBOARDS

EFFECTIVE VISUALIZATION PRINCIPLES

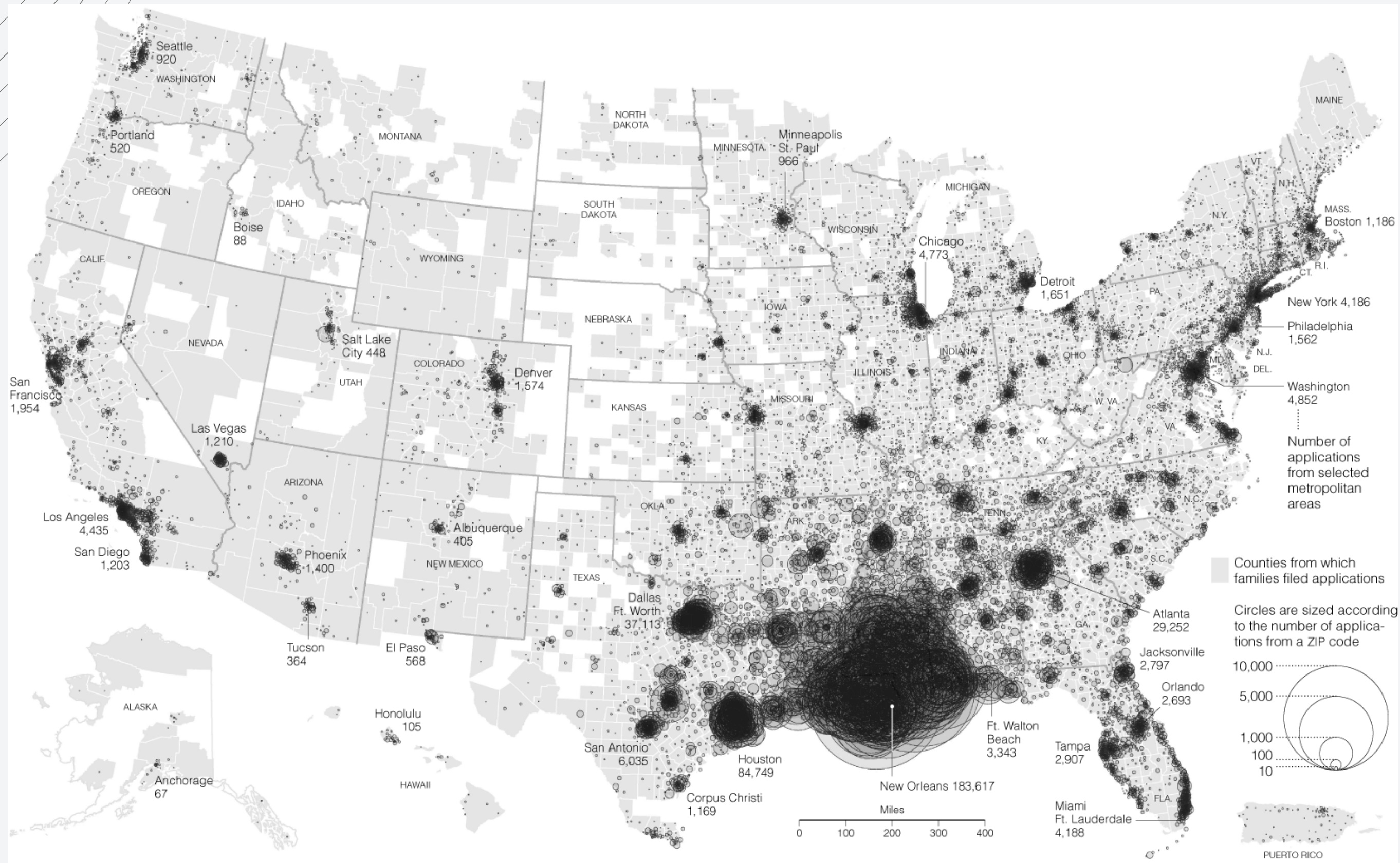
- *Clarity;*
- *Accuracy;*
- *Efficiency;*
- *Aesthetic appeal.*



Campanella then switched to an identically constructed map, only this time based on 2010 census data, and in bits and pieces on the screen there was a simple and arresting picture of what Katrina meant. In the neighborhoods that were once a dense black, many of the little squares had thinned and turned gray. The sharp lines that once separated the teapot from Central City were now blurry: the white areas of the city were pushing north, into the vacuum left by the exodus. The Bywater was graying, as it gentrified still further. “Before Katrina, an American Community Survey estimate of New Orleans Parish population was four hundred and fifty-five thousand, and about sixty-eight per cent black,” Campanella said. “Now the latest estimate is three hundred and eight-four thousand, and it’s about sixty per cent black. The white population has largely reconstituted itself numerically. So if you do that math, we’re talking about seventy-nine to eighty thousand fewer African-Americans.” In the most traumatic manner possible, Katrina caused what social scientists normally consider a good thing—a major move—and that is why, a decade after it devastated the Gulf Coast, the storm continues to defy simple categorization.

HURRICANE KATRINA

<https://www.newyorker.com/magazine/2015/08/24/starting-over-dept-of-social-studies-malcolm-gladwell>



HURRICANE KATRINA

<https://www.nytimes.com/interactive/2015/08/25/us/mapping-katrina-and-aftermath.html>

SOME STATISTICS

80%



Data produced since 2019

BIG DATA

Hal Varian

«The ability to take data – to be able to understand it, to process it, to extract value from it, to visualise it, to communicate it – is going to be a hugely important skill in the next decades.»



*Chief Economist, Google,
The McKinsey Quarterly, январь 2009*

REFERENCES

